

```
keep_going = False
if(iterations >= maximum_iterations):
    keep_going=False
iterations = iterations + 1
```

[Listing 3-2](#): An implementation of gradient ascent that climbs an income hill instead of a revenue hill

The code in [Listing 3-2](#) follows exactly the same gradient ascent algorithm as the revenue-maximization process we implemented previously. The only difference is the curve we are working with. Our taxation/revenue curve had one global maximum value that was also the only local maximum. Our education/income curve, by contrast, is more complicated: it has a global maximum, but also several local maximum values (local peaks or maxima) that are lower than the global maximum. We have to specify the derivative of this education/income curve (in the first lines of [Listing 3-2](#)), we have a different initial value (12.5 years of education instead of 70 percent taxation), and we have different names for the variables (`current_education` instead of `current_rate`). But these differences are superficial; fundamentally we are doing the same thing: taking small steps in the direction of the gradient toward a maximum until we reach an appropriate stopping point.

The outcome of this gradient ascent process is that we conclude that this person is overeducated, and actually about 12 years is the income-maximizing number of years of education. If we are naive and trust the gradient ascent algorithm too much, we might recommend that college freshmen drop out and join the workforce immediately to maximize earnings at this local maximum. This is a conclusion that some college students have come to in the past, as they see their high school–graduate friends making more money than them as they work toward an uncertain future. Obviously, this is not right: our gradient ascent process has found the top of a local hill, but not the global maximum. The gradient ascent process is depressingly local: it climbs only the hill it's on, and it isn't capable of taking temporary steps downward for the sake of eventually getting to another hill with a higher peak. There are some analogues to this in real life, as with people who fail to complete a university degree because it will prevent them from earning in the near term. They don't consider that